

Market Structure, Information, and Market Power in Farmland Transactions

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1 Introduction

Farmland markets operate in some of the thinnest and most localized competitive asset markets, where mostly only a few nearby landowners participate in each transaction and market power arises from a limited number of buyers and sellers (Cotteleer et al., 2008). Even though farmland collectively represents trillions of dollars in asset value, majority of farmland transactions involve minimal active participants. Parcels are usually bought by nearby farmers or operators, and the demand is shaped by geographical proximity, operational fit and farm-specific characteristics (Bigelow et al., 2020; Nickerson et al., 2012). These conditions stand in sharp contrast to the competitive assumptions that underlie much of the empirical and policy analysis in agricultural economics. In such settings, prices need not necessarily reflect its competitive fundamentals; instead, they may be shaped by the relative bargaining power of a small number of local buyers and sellers and by the information those participants possess about demand conditions, competition, and outside options in the market.

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Although thinness in farmland markets is widely acknowledged, we still know very little about how market power actually emerges in these environments. Most farmland literature uses individual transaction level data that reveal final prices, but they reveal almost nothing about the competitive structure that produced them, how many buyers seriously valued the parcel, whether multiple sellers were active in the area, what outside options each party had, how expectations and strategic behavior shaped the final outcome, or how private information shaped the negotiation behavior. Empirical studies have repeatedly shown that the local market structure is strongly correlated with farmland prices. Cotteleer et al. (2008) demonstrate that the rural Dutch land prices vary systematically with local ratio of buyers to sellers, and Kionka et al. (2024) provide experimental evidence that bargaining environments shape final outcomes in agricultural land markets, with local operators possessing more bargaining power than absentee or non-farming actors. However, these studies do not reveal why these patterns arise. Because the competitive landscape, buyer valuations, seller reservation prices, and bargaining dynamics are all unobserved in real-world data, empirical analyses cannot identify the micro-foundations of market power in farmland transactions.

A central challenge is that farmland markets are fundamentally local, frictional, and informationally asymmetric. The value of the parcel depends heavily on its proximity to existing fields of a farmer, field configuration, soil characteristics, and how well it fits the existing operation of the buyer (Guiling et al., 2009; Bigelow et al., 2020; Nickerson et al., 2012). These features create what is often referred to as local buyer scarcity: for many parcels, only a few nearby farmers have higher willingness to pay than anyone else. On the supply side of the market, only a few landowners typically bring land within a local market in a given year, which produces a potential seller scarcity. Together, these forces imply that many farmland sales often resemble small-N bargaining situations (Binmore et al., 1986; Wolfram, 1997), where prices reflect strategic interaction rather than competitive equilibrium.

At the same time, farmland transactions are characterized by significant information frictions. Buyers typically possess private information about how well a parcel fits their operation and how valuable it is relative to other parcels in sale. Sellers hold private information about their reser-

vation values and outside options. Neither side usually observes how many serious trading partners exist, how strongly other buyers value the parcel, or how competitive the local market truly is. These information frictions naturally interact with the market thinness, and jointly determine bargaining power and negotiated prices. These frictions: who values the land most, who has credible alternatives, and what each side knows about the market, are unobservable in administrative datasets, and the mechanism behind the market power in farmland transactions remains largely hidden.

This study uses a controlled laboratory experiment designed to replicate the core structural and informational features of real farmland market. In the experiment, buyers face an induced notion of "locality", where each buyer has a higher value for parcels that are "nearby", mimicking the way actual farmers value land close to their existing operations. Markets are intentionally thin, so each seller faces only a limited number of serious potential buyers. Participants receive private information about their assigned values and costs following the standard induced value theory (Smith, 1976). Prices are determined in a decentralized multilateral bargaining institution that reflects the private-treaty negotiation format common in farmland sales, in which buyers and sellers negotiate directly and outcomes depend on the availability of alternative trading partners rather than centralized price posting. The bargaining process follows experimental market institutions in which prices emerge from decentralized negotiation among multiple traders (Cason and Plott, 2005). This allows market power to arise endogenously through outside options, as in real farmland markets.

The experiment explores two fundamental dimensions of thinness. First, it varies local thinness by changing the number of high-value (neighbor) buyers for each parcel, thereby moving from bilateral bargaining environments to settings with buyer-side competition. The second dimension is global thinness, which is introduced by changing the number of buyers and sellers active in the market. Markets with relatively many sellers and few buyers give buyers more leverage, while markets with more buyers than sellers shift bargaining power on the seller side. Holding these structural features fixed, additional treatments vary information disclosure by revealing seller costs to buyers or demand conditions to sellers. Because the underlying distributions of

costs and values remain constant across treatments, any differences in outcomes, surplus division, or efficiency can be traced directly to changes in market structure or information availability.

This ability to vary structural features one at a time is the key strength of using controlled laboratory experiment, something that cannot be replicated with the farmland administrative transaction data. In real markets, we rarely know which buyers are actually active bidders, how many relevant sellers are in play, or how asymmetric information shape the bargaining process. The lab setting, by contrast, allows us to build a simplified but structurally accurate version of a farmland market, where these elements can be held constant or varied by design. This controlled environment makes it possible to see clearly when and why competitive predictions break down in thin, spatially segmented markets.

The goal of this paper is not to replicate every institutional nuance of real farmland transactions. Instead, the core idea is to isolate the fundamental price-formation mechanisms that arise from proximity-based valuations, small market size and informational friction. These forces appear to be central to bargaining outcomes in farmland markets, and the laboratory setting makes it possible to isolate them, something administrative data cannot do. By isolating these mechanisms, the experiment helps clarify when competitive benchmarks fail and how bargaining power develops in markets shaped by geography and limited participation. The insights extend beyond agricultural land market to other spatially constrained asset markets, such as housing, water rights, timber and mineral leases where geographical characteristics and small numbers play a similar role in determining prices and market power.

2 Conceptual Framework

Farmland transactions often occur in environments with few potential trading partners and highly heterogeneous valuations which are driven by spatial proximity, and information asymmetries. This section develops a framework that captures how these features jointly shape prices and the distribution of surplus in decentralized asset markets. The goal is not to derive a fully structural model of equilibrium bargaining, but rather to provide clear comparative statics that organize the

experimental treatments and generate testable predictions.

Consider a farmland market with a set of sellers $j \in \{1, \dots, S\}$, each holding one indivisible parcel, and a set of buyers $i \in \{1, \dots, B\}$, each with an unit demand. The seller-specific cost of each parcel is c_j , which is known only to seller j and represents the cost of private reservation value or opportunity cost of selling the parcel. Each Buyer i has a parcel-specific value v_{ij} , which reflect adjacency, field characteristics and other localized benefits that vary across buyers and parcels. If buyer i purchases parcel j at price p_j , the total match surplus is:

$$S_{ij} = v_{ij} - c_j. \quad (1)$$

Trade is efficient for rational buyers and sellers whenever $v_{ij} > c_j$ and both parties agree on a price $p_j \in [c_j, v_{ij}]$. The price determines how the total surplus $S_{ij} = v_{ij} - c_j$ is divided: the seller earns $\pi_j^S = p_j - c_j$ and the buyer earns $\pi_i^B = v_{ij} - p_j$. In a perfectly competitive market with many buyers and sellers and complete information, prices would be driven to sellers' costs ($p_j = c_j$), with buyers capturing all surplus. But competitive conditions rarely hold in farmland markets: the valuation of a parcel sharply differs across buyers, the set of serious buyers is often very small, and both costs and valuations are privately observed. These features generate bilateral bargaining power, so that the division of surplus between buyer and seller is determined by strategic interaction rather than competitive forces.

2.1 Bargaining with Outside Options

The negotiated price depends on the surplus available from trade and on what each party obtains if negotiations fail. Let $d_S \geq 0$ denote the seller's outside option, which is the payoff from trading with an alternative buyer, and $d_B \geq 0$ be the buyer's outside option, the payoff from trading with an alternative seller. The generalized Nash bargaining solution (Nash, 1950; Binmore et al., 1986) solves:

$$p_j^* = \arg \max_p \left(p - c_j - d_S \right)^\beta \left(v_{ij} - p - d_B \right)^{1-\beta}. \quad (2)$$

where $\beta \in [0, 1]$ is the seller's bargaining weight. The solution¹ is:

$$p_j^* = c_j + d_S + \beta(v_{ij} - c_j - d_S - d_B). \quad (3)$$

Prices increase in d_S and decrease in d_B : stronger outside options for the seller raise prices; stronger outside options for the buyer lower them. The competitive outcome $p_j^* = c_j$ emerges only when $d_B = v_{ij} - c_j$, requiring buyers to face abundant seller competition. In thin markets, outside options are weak on both sides and prices reflect strategic interaction rather than competitive forces. The experimental treatments shift d_S and d_B by varying the number of high-value buyers per parcel, the buyer-seller ratio, and information availability.

2.2 Local Thinness: Parcel-Specific Demand Concentration

Local thinness concerns the number of buyers who place a high value on a specific parcel. For each parcel j , if N_j denote the set of set of high-value buyers.

$$N_j = \{i : v_{ij} \text{ is high}\}, \text{ with size } n_j = |N_j|. \quad (4)$$

High-value buyers represent “neighbors”, for whom the the parcel generates strong localized benefits like proximity to existing operations, favorable field configuration, or operational efficiency. Remaining buyers have lower values because the parcel does not meaningfully enhance their existing operations.

The seller's outside option d_S increases in n_j : each additional high-value buyer strengthens the seller's position by providing a credible alternative trading partner. From equation 3, the prices increase in n_j .

Case 0: $n_j = 0$. No buyer values parcel highly. All buyers hold low valuations drawn from the same distribution, and the seller faces a pool of undifferentiated buyers with limited willingness to pay. Without any neighbor competition, the seller has no ability to leverage buyer rivalry. Prices are determined entirely through bilateral bargaining dynamics between the seller and individual

¹The solution follows from the first-order condition of equation 2.

low-value buyers, and the seller's bargaining power is weakest. This case defines the thinnest local environment and serves as the benchmark in the experiment.

Case 1: $n_j = 1$. Only one buyer values parcel highly and the seller effectively bargains with a single serious buyer. This resembles a bilateral bargaining environment, where both sides recognize that the other has few alternatives. Classic results in bargaining theory (Binmore et al., 1986) show that with limited outside options for both parties, prices deviate substantially from competitive benchmarks because each side has the ability to extract informational or strategic rents. Relative to Case 0, the presence of one high-value buyer shifts some bargaining power toward the seller.

Case 2: $n_j \geq 2$. Multiple buyers value the parcel highly. In this setting, each high-value buyer serves as an outside option for the seller, generating buyer-side competition. As n_j increases, d_s rises and from equation 3:

$$p_j \text{ increases in } n_j,$$

holding S and B constant. Together, Cases 0 through 2 imply that prices should be lowest when no neighbors are present, higher when one neighbor exists, and highest when multiple neighbors compete. Local thinness is therefore a parcel-level determinant of market power, and its effects are identified in the experiment by varying n_j across treatments while holding all other features of the market fixed.

2.3 Global Thinness: Market-Wide Buyer–Seller Ratios

Global thinness refers to the overall number of buyers and sellers in the market, independent of how values are distributed across parcels. Let the buyer-seller ratio is given by:

$$\theta = \frac{B}{S}. \tag{5}$$

Unlike local thinness, which operates at the parcel level through n_j , global thinness shifts the outside options of both sides simultaneously. When θ is large, that is, there are relatively more

buyers than sellers, buyers must compete for a limited supply of parcels. Each seller can credibly threaten to trade with another buyer, raising d_S . Buyers, facing scarce outside options, see d_B fall. From equation 3, both forces push prices upward.

Conversely, when θ is small, that is, there are relatively few buyers and many sellers in the market, sellers compete for limited demand. Each buyer can negotiate with multiple sellers simultaneously, raising d_B . Sellers face the risk of being left without a trading partner, weakening d_S . Both forces push prices downward. This is consistent with thin-market environments observed in Wolfram (1997) and in theoretical work on market concentration (Sutton, 1991). Formally:

$$p_j^* \text{ decreases as } \theta \text{ decreases,}$$

holding the distribution of v_{ij} and c_j constant. Global thinness therefore shifts the balance of surplus between buyers and sellers at the market level rather than the parcel level, and its effect is distinct from the local thinness channel.

2.4 Information Frictions

The local and global thinness analysis holds the information structure fixed: sellers observe only their own cost c_j and buyers observe only their own parcel-specific valuation v_{ij} . However, information availability in the farmland market may vary. Buyers may form accurate beliefs about the range of seller reservation values through county assessor records, extension publications, or regional land value benchmarks. Similarly, sellers may learn demand conditions through cash rent rates paid by neighboring operators, Farm Service Agency records, or observable land acquisition behavior of nearby farms. The degree to which each side can access such market-level information varies across transactions, making information availability an independently important determinant of bargaining power that operates separately from the physical structure of the market.

Private information distorts bargaining in two ways. It generates informational rents: the party with private information can exploit the other's uncertainty to extract surplus beyond what outside options alone would support. It also produces allocative inefficiency: Myerson and Satterth-

waite (1983) establish that under two-sided private information, no mechanism can simultaneously achieve efficient trade, incentive compatibility, and voluntary participation, so profitable trades are foregone in equilibrium.

The information channel is isolated by holding market structure fixed and varying what each side observes. Let $\mathcal{I}_S = \{c_j\}$ and $\mathcal{I}_B = \{v_{ij}\}$ denote the private information baseline.

Cost disclosure to buyers. The buyer's information set expands to $\mathcal{I}_B = \{v_{ij}, c_j\}$, reflecting environments where buyers have access to market-level information about seller reservation values through comparable sales data, county assessor records, or regional land value benchmarks. Knowledge of the seller's cost eliminates the seller's informational rent: the buyer can credibly anchor offers to c_j and reject prices that substantially exceed it, undermining the seller's ability to extract surplus through cost uncertainty. Prices fall relative to the private information baseline and surplus shifts toward buyers.

Demand disclosure to sellers. The seller's information set expands to $\mathcal{I}_S = \{c_j, n_j, \max_i v_{ij}\}$. Knowledge of the number of high-value buyers and the highest valuation eliminates the buyer's informational advantage. A seller who observes $n_j > 0$ can credibly maintain a high asking price knowing that at least one high-value buyer is present, undermining the buyer's ability to conceal a high valuation. Prices rise relative to the private information baseline and surplus shifts toward sellers.

Market thinness and information frictions therefore operate through distinct channels. Market thinness shifts outside options d_S and d_B by changing the market structure. Information frictions determine how much each side can exploit private knowledge of the other's type. Market structure and information availability are therefore separately identified sources of bargaining power, and their effects on prices and surplus division are different.

2.5 Interaction Effects

Local thinness, global thinness, and information frictions each operate through distinct channels, but their combined effects on prices and surplus division depend on how these forces interact.

Local and Global thinness interact in intuitive but important ways. A seller who faces only one high-value buyer ($n_j = 1$) may still secure a relatively high price if the global market is thick with buyers, because the buyer understands that the seller has abundant outside options. Conversely, a seller who attracts several high-value buyers ($n_j \geq 2$) for a given parcel may still end up accepting a low price if those buyers face plenty of alternative parcels in the market and therefore hold strong global outside options. Formally, the outside options d_S and d_B in equation (3) reflect both parcel-level and market-level competition simultaneously, so local and global thinness enter prices jointly.

Information plays a critical role in shaping these interactions. When buyers and sellers are uncertain about the number or strength of alternatives available to their counterparties, bargaining outcomes can differ substantially from competitive predictions. Sellers who believe demand is strong may hold out for higher prices, while buyers who believe competition is intense may concede surplus even when better terms are feasible. When information is disclosed, the informational rent is eliminated, and prices approach the complete-information benchmark even when market structure remains thin. In this sense, information transparency and market thickness act as substitutes: thicker markets erode the rents that arise from scarcity, while greater transparency erodes the rents that arise from asymmetric information, and either force can independently push prices toward competitive levels.

3 Experimental Design

The experiment implements a decentralized multilateral bargaining market which captures the core structural and informational features of farmland transactions: localized demand, heterogeneity in buyers' valuations across parcels, limited number of trading partners, and incomplete information about competition and outside options. Trade occurs through private negotiations between buyers and sellers, with prices determined through decentralized bargaining in most farmland transactions. Market power arises endogenously from participants' outside options, which are determined by local and global market thinness.

Each session includes 24 participants, divided into two independent subsessions of 12 participants. These subsessions function as entirely separate economies. Participants never interact across subsessions, and all randomization and treatments are implemented at the subsession level. Within each subsession, participants are assigned the role of Buyer or Seller at the beginning of the session. Each subsession contains two independent markets, and all interactions occur only within a participant's assigned market.

At the beginning of each period, every seller j in a market owns one indivisible “parcel of land” and privately observes a cost c_j drawn independently each round from $U[40, 50]$, which serves as the seller's reservation value or opportunity cost of selling the parcel. Similarly, each buyer i receives a value table listing a parcel-specific value v_{ij} for each seller's parcel in that market. These values captured the notion of “locality” that is central in farmland markets: for some parcels the buyer is a high-value “neighbor,” receiving a value v_{ij} drawn from $U[100, 110]$, while for other parcels the buyer has only a low value v_{ij} drawn from $U[65, 75]$

Trades occur through a decentralized multilateral bargaining agreement, formed during a fixed-length bargaining window. Buyers may send price offers to any seller, and sellers may send counteroffers to any buyer. Participants may negotiate simultaneously with multiple potential partners, creating multilateral outside options. A transaction occurs when an offer is accepted; once a transaction is completed, it is binding, the parcel is removed from the market for that period, and the buyer exits the market. Each seller can sell at most one parcel per period, and each buyer can purchase at most one parcel per period.

If buyer i purchases parcel j at price p_j , payoffs (in tokens) are given by:

$$\pi_i^B = v_{ij} - p_j, \quad (6)$$

$$\pi_j^S = p_j - c_j. \quad (7)$$

Buyers who do not purchase, and sellers who do not sell, earn zero in that period.

The baseline information environment is asymmetric. Sellers observe only their own cost c_j ,

and the offers they receive or send, but not buyers' valuations. Buyers, in turn, observe their own personal value of each parcel v_{ij} , and the offers they receive or send, but not sellers' costs and other buyers' valuations. Neither side observes the distribution of valuations or costs, nor how many buyers value any given parcel highly.

Information disclosure is varied across treatments by selectively revealing additional market information to buyers or sellers while holding costs and valuations fixed. These disclosures alter beliefs about outside options and competition without changing the underlying economic environment.

3.1 Treatments

The experimental treatments are designed to isolate two distinct sources of market power in decentralized asset markets: (i) the structure of trading opportunities, captured by local and global market thinness, and (ii) information frictions, captured by alternative disclosure regimes. Across all treatments, the distributions of seller costs and buyer valuations, the bargaining institution, and the matching structure are held constant. Only market structure or information availability is varied.

Market Structure Treatments: The first four treatments implement the comparative statics from the conceptual framework by holding the distributions of c_j and v_{ij} constant and varying only (i) the number of high-value buyers per parcel (local thinness) and (ii) the buyer-seller ratio (global thinness).

- **Treatment 1: Market Benchmark.** Each market has 2 sellers and 4 buyers. No buyer holds a high valuation for any parcel, all buyers draw values uniformly from $[65, 75]$. This configuration creates a locally thin environment in which sellers face no competition from high value neighbors, and must negotiate with undifferentiated buyers. This treatment serves as the benchmark against which the effects of increased local competition and global thinness are evaluated.

- **Treatment 2: One-Neighbor Local Thinness.** Market size remains 2 sellers and 4 buyers as in benchmark treatment, but each parcel now has exactly one high-value buyer and all the remaining buyers receive a low valuation in $v_{ij} \in \{65, 75\}$ for that parcel. This treatment creates a locally thin environment resembling bilateral bargaining at the parcel level. Any differences relative to Treatment 1 therefore isolate the effect of local thinness.
- **Treatment 3: Three-Neighbor Local Thinness.** Market size remains 2 sellers and 4 buyers. For each parcel, three buyers draw high valuations from $[100, 110]$, while the remaining buyer draws a low valuation from $[65, 75]$. This treatment creates strong buyer-side competition at the parcel level. Relative to Treatment 1, sellers can leverage the competition among multiple high-value buyers, and relative to Treatment 2, the additional neighbors further strengthen sellers' outside options.
- **Treatment 4: Global Thinness (Buyer Scarcity)** The buyer–seller ratio is reversed. Each market now has more sellers than buyers with 4 sellers and 2 buyers, while the underlying cost and value distributions remain the same. With more sellers than buyers, sellers must compete for a limited pool of buyers, and each buyer can negotiate simultaneously with multiple sellers. This gives buyers strong outside options even though their valuations are no higher than in Treatment 1. Sellers, by contrast, face the risk of being left without a trading partner if a buyer chooses a competitor. This shifts bargaining power toward buyers through market-wide scarcity rather than through any difference in parcel-level valuations. Relative to the benchmark, the only change is the buyer–seller ratio, so any differences in prices or surplus division can be attributed directly to global thinness.

These market structure treatments are visually shown in 1.

Information Disclosure Treatments: The remaining two treatments vary the information available to market participants while holding the market structure fixed at 2 sellers and 4 buyers. These treatments examine how information frictions interact with decentralized bargaining to shape market power.

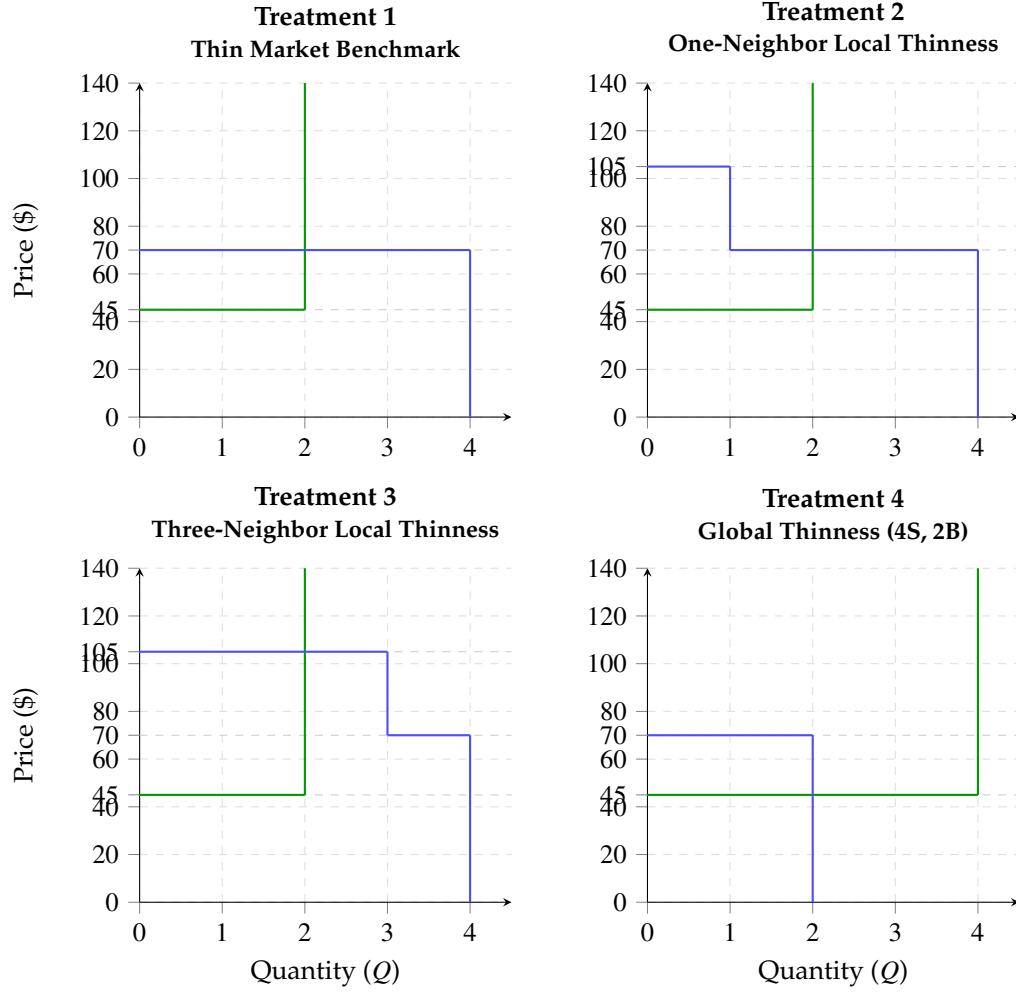


Figure 1: Induced Supply and Demand Schedules by Treatment

Notes: Supply and demand curves are drawn at distribution midpoints for illustration. Seller costs are drawn each round from $U[40, 50]$, low buyer valuations from $U[65, 75]$, and high buyer valuations from $U[100, 110]$. Treatments 1–3 hold the buyer–seller ratio fixed at 4:2 and vary the number of high-value buyers per parcel (0, 1, and 3 respectively). Treatment 4 reverses the ratio to 2:4 with all buyers holding low valuations, isolating the effect of global thinness.

- **Treatment 5: Cost Disclosure to Buyers** In addition to their own valuation tables, buyers observe the seller's cost c_j for each parcel in their market. Sellers continue to observe only their own cost. This treatment increases buyer-side information and is intended to strengthen

buyers' bargaining positions by making seller reservation values transparent.

- **Treatment 6: Demand Disclosure to Sellers** In addition to their own cost c_j , sellers observe information about demand for their parcel. Each seller in this treatment observes (i) the number of buyers in the market who hold a high valuation for their parcel, and (ii) the highest valuation among those buyers. Buyers do not observe any additional information. This treatment increases seller-side information and is intended to strengthen sellers' bargaining positions.

Together, these six treatments allow the experiment to separately identify the effects of market structure and information disclosure on prices, surplus division, and bargaining outcomes in decentralized asset markets. By varying only one dimension at a time, the design provides clean evidence on the sources of market power in environments that resemble farmland transactions and other thin asset markets.

3.2 Procedures

Each session includes 24 participants divided into four independent markets of six participants each. These markets operate as entirely separate economies, participants never interact across markets, and all randomization and treatments are implemented at the market level. Within each market, participants are randomly assigned the role of Buyer or Seller at the beginning of the session and retain that role for the first treatment block. Each session consists of two consecutive treatment blocks of 20 rounds each, for a total of 40 paid rounds per session.

The session schedule varies across days. On Day 1, participants complete Treatment 1 in rounds 1–20 and Treatment 4 in rounds 21–40. At the start of round 21, roles are reversed: participants who served as Buyers in Treatment 1 become Sellers, and those who were Sellers become Buyers. On Day 2, participants complete Treatment 2 in rounds 1–20 followed by Treatment 3 in rounds 21–40, with roles held fixed throughout. On Day 3, participants complete Treatment 5 in rounds 1–20 followed by Treatment 6 in rounds 21–40, again with roles fixed. A different group of 24 participants is recruited for each day, so the design is between-subject across treatment pairs. At

the transition between treatments, participants receive an on-screen notification describing their role for the upcoming block and the relevant timing rules. No verbal communication between participants or between participants and the experimenter takes place during the session. All instructions and feedback are delivered through the computer interface.

At the beginning of each round, sellers privately observe their cost c_j and buyers privately observe their parcel-specific valuations v_{ij} , both drawn independently from the distributions described above. Sellers then post an asking price, after which a negotiation phase begins. The negotiation phase lasts 90 seconds for the first five rounds of each treatment block and 60 seconds for rounds 6–20 of each block.² The round ends when time expires or when no further trades are possible in the market. At the end of each round, participants see a results screen displaying whether a trade occurred, the transaction price if trade occurred, and their earnings for the round. Participants do not observe other participants' costs, valuations, or earnings at any point.

At the end of the session, four rounds are randomly selected from those in which the participant earned a positive profit. Earnings from those rounds are converted to cash at a rate of 1 token = 0.025, and all participants receive a \$10 show-up fee regardless of performance. Payment is made privately at the conclusion of the session. The random selection of paying rounds reduces wealth effects and encourages participants to treat each round as an independent decision. All sessions are conducted using oTree (Chen et al., 2016).

4 Hypothesis

The conceptual framework yields a set of predictions about how prices and surplus should respond to the structure of trading opportunities and the availability of information. These predictions guide the analysis of the experimental data. Because the induced cost and value distributions are held constant across treatments, any systematic differences in outcomes can be attributed directly to the changes in local thinness (the number of high-value buyers per parcel), global thinness (the buyer–seller ratio), or the information available to buyers and sellers about market conditions.

²The longer window in early rounds allows participants to become familiar with the interface and bargaining process.

- **H1: Prices increase with the number of high-value buyers per parcel (Local Competition Effect).** In a market where no buyers hold high valuation for a parcel, the seller faces a pool of undifferentiated buyers. Introducing one high-value buyer creates a single serious trading partner, which shifts some bargaining power toward the seller. Adding more high-value buyers creates buyer-side competition, further strengthening the seller's outside options. Consequently, final prices should be lowest in the no-neighbor benchmark, higher in the one-neighbor environment, and highest in the three-neighbor environment. Formally, if p_j^k denote the average price in a treatment with k high-value buyers per parcel, the prediction is:

$$p_j^0 < p_j^1 < p_j^3.$$

This prediction follows directly from the idea that sellers' bargaining power increases with the number of serious buyers competing for a parcel.

- **H2: Prices decrease when buyers are scarce relative to sellers (Global Thinness Effect).** When the buyer-seller ratio shifts from 4:2 (benchmark) to 2:4 (Treatment 4), sellers face more intense competition for a limited number of buyers. Buyers, in turn, hold strong outside options as they can negotiate with multiple sellers simultaneously. As a result, negotiated prices are expected to be substantially lower in the global thinness treatment relative to the benchmark, even though the underlying distributions of costs and values are identical.
- **H3: Surplus allocation shifts with bargaining power.** Total surplus from a buyer-seller match is given by $v_{ij} - c_j$, and the distribution of surplus between the two sides depends on the strength of each side's outside options. When local competition favors sellers, as in Treatments 2 and 3 relative to the benchmark, sellers have stronger alternatives and should capture a larger share of the surplus. When global thinness favors buyers, as in Treatment 4, buyers should claim a larger share.
- **H4: Cost disclosure to buyers lowers prices and shifts surplus toward buyers.** When buyers observe sellers' costs, they gain information about sellers' reservation values. This weak-

ens sellers' ability to extract surplus through bargaining. Buyers can change their offers based on sellers' costs and credibly avoid prices far above the seller's reservation value. As a result, negotiated prices should be lower and buyer surplus shares higher under cost disclosure than under the baseline private-information environment, holding market structure fixed.

- **H5: Demand disclosure to sellers raises prices and shifts surplus toward sellers.** When sellers observe the the number of buyers with high valuations and the highest valuation among them, they gain information about the demand for their asset. This information strengthens sellers' bargaining position during the price negotiation. Negotiated prices should therefore increase and seller surplus shares should rise relative to the baseline private-information environment.

Participant Instructions

Overview

Welcome to the market experiment. You are about to participate in a market experiment. Your earnings will depend on your own choices, the choices of other participants, and the prices set in the market. All interaction is anonymous. If you have a question at any time, please raise your hand and an experimenter will come to your place and assist you.

The experiment consists of multiple rounds. In each round, you will interact with a small group of participants in a market. In this market, some participants will be Sellers and others Buyers, which is randomly assigned and your role will be displayed on the computer screen in the beginning of each sessions. Sellers may sell at most one unit per round, and Buyers may purchase at most one unit per round.

All decisions in this experiment are made anonymously. The experiment includes 2 unpaid practice rounds and then 20 paid rounds. At the end of the experiment, the computer will calculate your earnings from all 20 paid rounds and select your **four random periods from the paid rounds**. Tokens from these rounds are converted to cash at the rate of 1token = \$0.25. You will also receive a **\$10 show-up payment**.

Groups and Roles

You will be randomly assigned to a **group** of participants at the beginning of the session. You will remain in the same group for the duration of the session. Within your group, you will interact only with participants assigned to your market. You will never interact with participants from other groups.

At the start of each **treatment**, you will be randomly assigned a **role**:

- **Sellers**, who may offer one unit for sale each round.
- **Buyers**, who may attempt to buy one unit per round.

Your role will remain the same for all rounds *within* a session. If a new session begins, your role may be reassigned, and your current role will be displayed on the computer screen when a new session begins. Your screen will also show how many Buyers and Sellers are in your market.

What Sellers See

At the beginning of each round, each Seller receives one unit that can potentially be sold in that round. Each Seller is privately assigned a cost for the round. The cost is private information: only that Seller sees it.

The cost represents how many tokens the Seller effectively gives up by selling the unit. If the unit is not sold, the Seller keeps the unit and earns 0 tokens for the round.

If a Seller and a Buyer agree on a transaction price p , then the Seller's earnings are:

$$\pi^S = p - c.$$

Example. If your cost is 50 and you sell the unit for 80, your earnings are:

$$80 - 50 = 30 \text{ tokens.}$$

If your unit remains unsold, your earnings for that round are:

$$0 \text{ tokens,}$$

since unsold units generate no profit.

In some parts of the experiment, Sellers may also see limited additional information about demand for their unit. Any such information will be clearly displayed on your screen.

What Buyers See

At the beginning of each round, each Buyer sees a table showing their **value** for each Seller's unit. These values may differ across Sellers and may change from round to round. Only you see your own values.

If you buy Seller j 's unit at price p_j , your earnings for the round are:

$$\pi^B = v_j - p_j.$$

Examples. If your value for Seller A is 110:

- If you pay 80, you earn $110 - 80 = 30$ tokens.
- If you pay 90, you earn $110 - 90 = 20$ tokens.
- If you pay 120, you earn $110 - 120 = -10$ tokens.

If you choose not to buy any unit, your earnings for that round are 0 tokens.

In some parts of the experiment, Buyers may also see additional information about Sellers. Any such information will be clearly displayed on your screen.

Sequence of Decisions in Each Round

Each round follows the same sequence.

- **Stage 1: Negotiation Phase:** At the beginning of the round, Sellers observe their costs and Buyers observe their values. The market then enters a **90-second negotiation phase**.

During this phase, all participants may engage in negotiations. Sellers may make price offers, Buyers may accept offers, reject offers, or make counteroffers to Sellers. All offers and counteroffers must be whole numbers within the price range shown on the screen.

Offers and counteroffers are not binding unless they are accepted. Participants may make and respond to offers at any time during the 90 seconds.

Each Seller can sell **at most one** unit per round, and each Buyer can purchase **at most one** unit per round. Once a Seller's unit is sold, all remaining negotiations involving that unit end.

If no agreement is reached before the 90-second negotiation period expires, no transaction occurs for that Seller or Buyer in that round.

- **Stage 2: Feedback Phase:** At the end of the round, participants receive feedback on the screen. Sellers see whether their unit was sold, the agreed transaction price (if any), and their earnings for the round. Buyers see whether they purchased a unit, the price paid, and their earnings for the round.

Participants do not observe other participants' costs, values, or earnings.

Earnings and Payment

You will begin with 2 unpaid practice rounds to become familiar with the interface. After the practice rounds, there will be **20 paid rounds** in your treatment. At the end of the experiment, the computer will select four random periods from the paid rounds, and earnings from those periods are converted to cash at a known exchange rate. Your token earnings from these four rounds will be summed and converted into cash at the announced rate. All payments are made privately.

Important Rules

- No talking or communication with other participants.
- All decisions must be made only through the computer interface.
- If you have any questions, please raise your hand.

Please make your decisions carefully, as they directly affect your earnings.

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